

Dear Editors,

Please consider my manuscript, “Observation-conditioned bounds define experimental decision criteria for bifunctional enzyme systems,” for publication as a Research Article in *PLOS Computational Biology*. It addresses a practical problem in mechanistic systems biology: how an experiment can support a quantitative decision when distinct molecular explanations reproduce the same observed response. The central contribution is a conservation-aware framework that derives measurable conditions for class-wide prediction guarantees and identifies complementary observations when mechanism discrimination is required.

The study makes three linked contributions. First, it constructs nitrogen-signaling mechanisms that preserve complete stationary target distributions, then derives a concurrent-count condition under which a proposed GlnE concentration meets a prescribed discrepancy tolerance. Second, a general graph enclosure extends the calculation beyond exact enzyme-concentration cancellation. Its application to AceK–isocitrate dehydrogenase retains two-enzyme-bound target complexes, while published wild-type and mutant kinetic summaries supply a separate test of binding ambiguity and independent-activity constraints. Third, a raw-assay benchmark includes nonlinear fitting, parametric null calibration and thousands of independent trials per evaluation setting. Condition-specific signal fits correct tested background and gain distortions; hidden affinity mixtures still undermine a single-class mechanistic interpretation.

The advance is the connection between a declared biochemical model class, an observable gate and an experimental decision. It complements prediction profile likelihood and model-discrimination design by providing sufficient conditions that do not require selecting one microscopic mechanism. The two enzyme systems and reusable graph algorithm support its relevance to the broad systems-biology readership of *PLOS Computational Biology*.

The study is deliberately model-class conditional. The manuscript distinguishes source-summary calibration, independent biochemical checks and proposed measurements; neither worked concentration criterion is presented as already certified in a native assay. Executable analyses, source provenance, complete trial records and independent numerical verification accompany the submission.

If available, I suggest Julio R. Banga (Computational Biology Lab, MBG-CSIC) as Academic Editor because of his expertise in mechanistic systems biology, model identifiability and experimental design.

This manuscript is original, has not been published, and is not under consideration elsewhere. I am the sole author and declare no competing interests. Reused data and software are cited, with provenance and licenses documented.

Thank you for considering the manuscript.

Sincerely,

Yule Choi

Korea Science Academy of KAIST

Busan, Republic of Korea

25-117@ksa.hs.kr